Piper PA-28R-201 Arrow III/IV (ARROW)

Lycoming IO-360-C1C6, 200 hp · 4 seats · Typical Arrow III (1977+) / Arrow IV POH procedures; fuel-injected, constant-speed prop, retractable gear, 2,750 lb MTOW. Automatic gear extension system where still fitted.

V-speeds & limits

Vr	**60–70 KIAS**
Vx	**78 KIAS**
Vy	**90 KIAS**
Best glide	**79 KIAS**
Vfe	**103 KIAS**
Vle	**129 KIAS**
Vlo	**107 KIAS retract / 129 KIAS extend**
Va	**118 KIAS (2,750 lb) / 96 KIAS (1,801 lb)**
Vno	**146 KIAS**
Vne	**183 KIAS**
Vs / Vso	**60 / 55 KIAS**
Approach	**75 KIAS (flaps 40°); 72 KIAS short field**

Normal procedures

Before Start

Preflight inspection	**COMPLETE**
Fuel quantity	**CHECK visually (36 gal usable per tank, 72 total)**
Baggage door	**CLOSED & LOCKED**
Passenger brief	**COMPLETE (belts, door, exits, sterile cockpit)**
Seats, belts, harnesses	**ADJUST & LOCK**
Door	**CLOSED & LATCHED (top and side)**
Parking brake	**SET**
Gear selector	**DOWN**
Emergency gear lever	**NEUTRAL (not in override)**
Avionics master	**OFF**
Electrical switches	**OFF**
Circuit breakers	**CHECK IN**
Alternate static source	**OFF (closed)**
Fuel selector	**FULLER TANK**
Propeller control	**FULL FORWARD (high RPM)**
Alternate air	**OFF (closed)**

Engine Start

Master switch	**ON**
Gear lights	**THREE GREEN**
Beacon	**ON**
Throttle	**FULL OPEN (prime)**
Mixture	**FULL RICH**
Electric fuel pump	**ON until fuel flow indicates, then OFF**
Mixture	**IDLE CUT-OFF**
Throttle	**OPEN 1/2 inch**
Propeller area	**CLEAR**
Starter	**ENGAGE (10 s max, then 30 s cool-down)**
Mixture	**ADVANCE to FULL RICH as engine fires**
Throttle	**ADJUST 800–1,000 RPM**
Oil pressure	**CHECK rising within 30 s (55–95 psi green)**
Ammeter / alternator	**CHECK charging**
Electric fuel pump	**OFF (fuel pressure check 14–45 psi)**
Avionics master	**ON**

Hot start: throttle 1/2 inch, mixture idle cut-off, pump OFF; advance mixture as engine fires. Flooded: throttle full, mixture idle cut-off, crank; retard throttle and advance mixture as engine fires.

Before Taxi

Flight instruments / avionics	**CHECK & SET**
Altimeter	**SET**
Heading indicator	**SET to compass**
Transponder	**STBY / as required**
Parking brake	**RELEASE**
Brakes	**CHECK during initial roll**
Steering	**CHECK (nosewheel steering with rudder pedals)**
Gyros	**CHECK during turns (turn coordinator, DG, attitude)**

Before Takeoff

Parking brake	**SET**
Flight controls	**FREE & CORRECT**
Fuel selector	**FULLER TANK**
Mixture	**FULL RICH**
Propeller control	**FULL FORWARD**
Throttle	**2,000 RPM**
Magnetos	**CHECK (max drop 175 RPM, max difference 50 RPM)**
Propeller	**CYCLE 2–3 times (RPM drop approx 300–500, oil pressure steady), then FULL FORWARD**
Alternate air	**CHECK (slight RPM drop), then OFF**
Vacuum	**4.8–5.2 in Hg**
Oil pressure / temperature	**CHECK GREEN (55–95 psi)**
Fuel pressure	**CHECK GREEN (14–45 psi)**
Ammeter	**CHECK charging**
Annunciator panel	**PRESS TO TEST**
Throttle	**IDLE check (500–700 RPM), then 1,000 RPM**
Electric fuel pump	**ON**
Flaps	**SET (0° normal / 25° short field)**
Trim	**SET for takeoff (neutral)**
Gear lights	**THREE GREEN, gear selector DOWN**
Emergency gear lever	**NEUTRAL (override only if briefed)**
Seat belts / harnesses	**FASTENED**
Door	**LATCHED (top and side)**
Pitot heat / lights	**AS REQUIRED**
Transponder	**ALT**
Takeoff briefing	**COMPLETE (abort point, EFATO plan, gear-up point)**

After Takeoff / Climb

Airspeed	**90 KIAS (Vy) / 78 KIAS (Vx, obstacles)**
Landing gear	**UP (positive rate, below 107 KIAS, no usable runway)**
Gear lights	**OUT, gear-unsafe light OUT**
Flaps	**RETRACT (if used, when clear of obstacles)**
Electric fuel pump	**OFF (at safe altitude, check fuel pressure)**
Power	**FULL THROTTLE / 2,700 RPM (25 in Hg / 2,500 RPM cruise climb at 104 KIAS)**
Mixture	**FULL RICH (lean above 5,000 ft)**
Engine instruments	**CHECK**
Landing light	**OFF (as desired)**

Cruise

Power	**SET (2,300–2,500 RPM, manifold pressure per chart, 75% max)**
Mixture	**LEAN (per power chart / fuel flow)**
Fuel selector	**CHANGE TANKS every 30–60 min (pump ON during change)**
Engine instruments	**CHECK**
Fuel quantity	**CHECK against plan**

Descent

Altimeter	**SET**
Mixture	**ENRICHEN as altitude decreases**
Power	**REDUCE gradually (keep manifold pressure above approx 15 in Hg)**
Fuel selector	**FULLER TANK**
Airspeed	**BELOW 129 KIAS before gear extension**
Approach briefing	**COMPLETE**

Before Landing

Seat belts / harnesses	**FASTENED**
Fuel selector	**FULLER TANK**
Electric fuel pump	**ON**
Mixture	**FULL RICH**
Propeller control	**FULL FORWARD (on final)**
Landing gear	**DOWN (below 129 KIAS)**
Gear lights	**THREE GREEN, nose gear in mirror**
Flaps	**SET (below 103 KIAS; 40° normal landing)**
Airspeed	**75 KIAS final (flaps 40°)**
Landing light	**ON**
Gear	**RECHECK THREE GREEN on final**

After Landing

Flaps	**RETRACT**
Electric fuel pump	**OFF**
Transponder	**STBY / as required**
Lights	**AS REQUIRED**

Shutdown

Parking brake	**SET**
Avionics master	**OFF**
Electrical switches	**OFF**
Propeller control	**FULL FORWARD**
Throttle	**1,000 RPM**
Mixture	**IDLE CUT-OFF**
Magnetos	**OFF (after prop stops)**
Master switch	**OFF**
Fuel selector	**OFF (if parking overnight / as per operator)**
Control lock / chocks	**INSTALL**

Abnormal procedures

Gear Fails to Retract / Gear Unsafe Light

Red gear-unsafe light or gear horn after selecting UP

Airspeed	**BELOW 107 KIAS (Vlo retract)**
Emergency gear lever	**CHECK NEUTRAL (lever in override prevents normal auto operation; auto-extension system lowers gear below approx 95–105 KIAS at low power)**
Gear selector	**RECYCLE DOWN, three green, then UP**

▶ Gear retracts, lights out?

Yes:

Flight	**CONTINUE, monitor on next extension**

No:

Gear selector	**DOWN**
Gear lights	**THREE GREEN**
Airspeed	**BELOW 129 KIAS**
Deferred: Gear left down – limit speed to 129 KIAS, reduced climb and range. Land as soon as practical.	**NOTED**

Gear Warning Horn

Horn with gear up: throttle below approx 14 in Hg, or auto-extension speed, or flaps beyond 10°

Gear position ..... **CHECK**

▶ Intentional gear-up (e.g. descent, override)?

Yes:

- Horn ..... **EXPECT; extend gear before landing**
- Gear ..... **DOWN before final, three green**

No – landing:

- Landing gear ..... **DOWN, three green**
- Emergency gear lever ..... **CHECK NEUTRAL**

Alternator Failure

Ammeter zero / discharge, ALT annunciator

Ammeter ..... **CHECK (zero output)**

ALT circuit breaker ..... **CHECK, RESET once**

ALT switch ..... **OFF, then ON**

▶ Alternator restored?

Yes:

- Ammeter ..... **MONITOR**

No:

- ALT switch ..... **OFF**
- Electrical load ..... **REDUCE to minimum (one radio, transponder)**
- Battery ..... **CONSERVE for gear extension (approx 30 min)**
- Landing gear ..... **EXTEND early; use emergency lever if battery is depleted**

Deferred: Alternator failed – limited battery; extend gear early (free-fall if needed), expect no gear lights / electric flaps late in flight. ..... **NOTED**

Low Oil Pressure

Indication below 25 psi at idle / below 55 psi in flight

Oil pressure / temperature ..... **CROSS-CHECK**

▶ Oil temperature rising with low pressure?

Yes – real:

- Power ..... **REDUCE to minimum required**
- Landing site ..... **SELECT, prepare for engine failure (prop may not respond to control)**

Deferred: Low oil pressure – land as soon as possible, expect engine failure and loss of prop governing. ..... **NOTED**

No – gauge suspect:

- Flight ..... **CONTINUE to nearest suitable airport, monitor**

Deferred: Oil pressure indication unreliable – land as soon as practical. ..... **NOTED**

High Oil Temperature

Indication above 245°F / in the red

Mixture ..... **ENRICHEN**

Power ..... **REDUCE**

Airspeed ..... **INCREASE (lower nose)**

Oil pressure ..... **CHECK**

▶ Temperature decreasing?

Yes:

- Flight ..... **CONTINUE, monitor**

No:

Deferred: Engine overheat – land as soon as possible, expect possible engine failure. ..... **NOTED**

Rough Running / Loss of Power

Rough engine, fluctuating fuel flow or RPM

Electric fuel pump ..... **ON**

Fuel selector ..... **SWITCH TANKS**

Mixture ..... **ADJUST for smoothest running**

Alternate air ..... **ON (induction icing / blockage)**

Magnetos ..... **CHECK L / R / BOTH (operate on best if one is faulty)**

▶ Engine smooth again?

Yes:

Settings ..... **LEAVE as found**

Deferred: Rough running cleared – pump on / alternate air on; land as soon as practical and investigate. ..... **NOTED**

No:

Power ..... **REDUCE to smoothest setting**

Deferred: Rough-running engine – land as soon as possible; expect engine failure. ..... **NOTED**

Low Fuel Pressure / Fuel Flow

Fuel pressure below 14 psi or fluctuating fuel flow

Electric fuel pump ..... **ON**

Fuel selector ..... **SWITCH to fuller tank**

Mixture ..... **FULL RICH**

▶ Pressure restored?

Yes:

Electric fuel pump ..... **LEAVE ON**

Deferred: Electric fuel pump on continuously – engine-driven pump suspect. Land as soon as practical. ..... **NOTED**

No:

Fuel exhaustion ..... **CONSIDER**

Prepare for engine failure – see Engine Failure in Flight.

Propeller Overspeed

RPM above 2,700 and not responding to prop control

Throttle ..... **RETARD**

Oil pressure ..... **CHECK**

Propeller control ..... **FULL DECREASE RPM, then set if any control remains**

Airspeed ..... **REDUCE**

Throttle ..... **AS REQUIRED to keep RPM at or below 2,700**

Deferred: Prop governor failed – control RPM with throttle and airspeed; land as soon as practical. ..... **NOTED**

Flap Failure

Flap circuit breaker ..... **CHECK, RESET once**

Flap switch ..... **RECYCLE**

▶ Flaps operable?

Yes:

Flaps ..... **AS REQUIRED**

No:

Approach ..... **80–85 KIAS, flaps-up landing**

Deferred: Flaps inoperative – flaps-up landing at 80–85 KIAS, longer landing distance. ..... **NOTED**

Door Open in Flight

Airspeed ..... **SLOW to approx 89 KIAS**

Storm window ..... **OPEN (cabin vent / overhead vent closed)**

Door ..... **PULL CLOSED, latch side then top**

▶ Door latched?

Yes:

Flight ..... **CONTINUE**

No:

Aircraft is controllable with door unlatched. Expect noise and some yaw.

Deferred: Door unlatched – land normally, secure door on the ground. ..... **NOTED**

Vacuum Failure

Vacuum below 4.8 in Hg / gyros drifting

Vacuum gauge	CHECK
Attitude indicator / DG	CROSS-CHECK against turn coordinator and compass, then COVER
IMC	EXIT, request no-gyro assistance if needed

Deferred: Vacuum failed – no attitude indicator / DG. Avoid IMC; partial-panel landing. — NOTED

Static Source Blocked

Airspeed / altimeter / VSI erratic

Pitot heat	ON
Alternate static source	OPEN (valve under left side of panel)
Storm window / vents	CLOSED (for calibration accuracy)

Airspeed and altimeter read slightly high with alternate static; see POH correction table.

Deferred: Alternate static in use – small airspeed/altimeter errors; gear/flap speeds by indication. — NOTED

Inadvertent Icing

Pitot heat	ON
Alternate air	ON
Cabin heat / defroster	ON (windshield)
Exit icing conditions	TURN / CLIMB / DESCEND
Airspeed	INCREASE, avoid abrupt maneuvers
Flaps	LIMIT (flaps-up or 10° landing with residual ice)

Deferred: Residual airframe ice – flaps-up / 10° landing at increased speed, longer runway; extend gear early. — NOTED

Emergency procedures

Engine Failure During Takeoff Roll

Loss of power before rotation

Throttle (memory)	CLOSED
Brakes (memory)	APPLY, stop straight ahead
Mixture	IDLE CUT-OFF
Magnetos	OFF
Master switch	OFF
Fuel selector	OFF

Engine Failure After Takeoff

Airborne, insufficient runway remaining

LAND IMMEDIATELY

Airspeed (memory)	79 KIAS (best glide)
Landing area (memory)	SELECT straight ahead (small heading changes only)

▶ Altitude and time available for a restart attempt?

Yes:

Fuel selector	SWITCH to other tank
Electric fuel pump	CHECK ON
Mixture	FULL RICH
Alternate air	ON

No – commit to landing:

▶ Landing surface?

Prepared – gear down:

Landing gear	DOWN (emergency lever if needed)

Rough / water – gear up:

Landing gear	UP
Emergency gear lever	OVERRIDE (prevent auto-extension)
Flaps	AS REQUIRED
Fuel selector	OFF
Mixture	IDLE CUT-OFF
Magnetos	OFF
Master switch	OFF (before touchdown)
Door	UNLATCH before touchdown

Do not attempt to turn back to the runway at low altitude.

Engine Failure in Flight

Loss of power with altitude available

LAND AS SOON AS POSSIBLE

Airspeed (memory)	79 KIAS (best glide)
Propeller control (memory)	FULL AFT (low RPM, extends glide)
Fuel selector (memory)	SWITCH to tank containing fuel
Electric fuel pump (memory)	ON
Mixture (memory)	FULL RICH
Alternate air (memory)	ON
Emergency gear lever	OVERRIDE (prevent auto-extension while gliding)
Engine gauges	CHECK for cause (fuel pressure, fuel flow, oil, RPM)

▶ Propeller stopped (not windmilling)?

Yes – crank:

Magnetos	BOTH
Throttle	OPEN 1/2 inch
Starter	ENGAGE

No – windmilling:

Magnetos	CHECK BOTH (try L / R individually)
Throttle	CYCLE, then set as required
Mixture	LEAN slightly if flooded

▶ Engine restarted?

Yes:

Propeller control	AS REQUIRED
Power	REDUCE to minimum required
Engine instruments	MONITOR
Deferred: Engine restarted after in-flight failure – land as soon as possible, suspect fuel system or induction cause.	NOTED

No – forced landing:

Forced landing without power:

Landing area	SELECT, plan approach into wind
Transponder / radio	7700, MAYDAY (position, intentions)
Seat belts / harnesses	SECURE

▶ Landing surface?

Prepared – gear down:

Landing gear	DOWN (free-fall with emergency lever if needed), three green

Rough / water – gear up:

Landing gear	UP, emergency lever in OVERRIDE
Flaps	AS REQUIRED (full on final)
Airspeed	75 KIAS final (flaps 40°)
Fuel selector	OFF
Mixture	IDLE CUT-OFF
Magnetos	OFF
Master switch	OFF (when no longer required)
Door	UNLATCH before touchdown

Engine Fire in Flight

LAND IMMEDIATELY

Fuel selector (memory)	OFF
Throttle (memory)	CLOSED
Mixture (memory)	IDLE CUT-OFF
Electric fuel pump (memory)	CHECK OFF
Cabin heat / defroster (memory)	OFF
Airspeed	INCREASE to blow out fire (do not exceed 146 KIAS)
Propeller control	FULL AFT once fire is out
Forced landing	EXECUTE (do not attempt restart)
Landing gear	DOWN if surface permits (emergency lever)
Master switch	OFF before touchdown

Engine Fire During Start

- Starter (memory) ..... **CONTINUE CRANKING (draw fire into engine)**
- Mixture (memory) ..... **IDLE CUT-OFF**
- Throttle (memory) ..... **FULL OPEN**
- Electric fuel pump ..... **OFF**
- Fuel selector ..... **OFF**

▶ Fire continues?

Yes:

- Magnetos / master ..... **OFF**
- Aircraft ..... **EVACUATE, fire extinguisher**

No:

- Engine ..... **SHUT DOWN, inspect before further flight**

Electrical / Cabin Fire

LAND AS SOON AS POSSIBLE

- Master switch (memory) ..... **OFF**
- Avionics master (memory) ..... **OFF**
- Cabin heat / defroster (memory) ..... **OFF**
- Vents ..... **OPEN**
- Fire extinguisher ..... **USE (if fire in cabin)**

After discharge, ventilate the cabin before landing.

▶ Fire out and electrical power essential?

Yes:

- Circuit breakers ..... **CHECK, do not reset tripped breakers**
- Master switch ..... **ON**
- Electrical items ..... **ON one at a time, watch for smoke**
- Deferred: Electrical fire – faulted circuit isolated. Verify gear and flap operation; use emergency gear extension if needed. ..... **NOTED**

No:

Master off: gear must be extended with the emergency gear lever (free-fall); no gear lights, no electric flaps.

- Deferred: Master off – emergency (free-fall) gear extension, flaps-up landing, no gear lights. ..... **NOTED**

Emergency Descent

- Throttle (memory) ..... **CLOSED**
- Propeller control (memory) ..... **FULL FORWARD**
- Mixture (memory) ..... **FULL RICH**
- Landing gear (memory) ..... **DOWN (below 129 KIAS)**
- Flaps (memory) ..... **UP**
- Airspeed (memory) ..... **129 KIAS (gear down) / 118 KIAS turbulent**

Emergency Gear Extension

Gear fails to extend with selector, or no three green

- Master switch — **ON**
- Gear circuit breaker — **CHECK IN**
- Panel lights — **OFF (daytime, to check gear lights)**
- Gear light bulbs — **CHECK (swap bulbs if one is out)**
- Airspeed — **BELOW 87 KIAS**
- Gear selector — **DOWN**
- Emergency gear lever — **DOWN (emergency down / free-fall position), hold**

▶ Three green?

Yes:

- Emergency gear lever — **LEAVE DOWN, do not retract**
- Deferred: Gear extended by free-fall – leave emergency lever down, do not retract gear, normal landing. — **NOTED**

No:

- Aircraft — **YAW side to side (rudder) to help gear lock**
- Airspeed — **REDUCE further (approach stall margin)**

▶ Three green now?

Yes:

- Deferred: Gear extended by free-fall after yawing – leave emergency lever down, normal landing. — **NOTED**

No:

Proceed to Gear-Up / Partial Gear Landing.

Gear-Up / Partial Gear Landing

Gear cannot be extended or locked

- Runway — **LONGEST available, emergency services notified (soft surface not required)**
- Fuel — **BURN DOWN to minimum practical**
- Seat belts / harnesses — **TIGHT**
- Loose items — **SECURE**

▶ Gear condition?

All up:

- Gear selector — **UP**
- Emergency gear lever — **OVERRIDE (prevent auto-extension)**

Partial – one or two legs:

- Gear — **RETRACT if possible; otherwise land on extended wheels, hold unsafe side off with aileron**
- Approach — **NORMAL, flaps 40°, 75 KIAS, power to touchdown**
- Electric fuel pump — **OFF**
- Mixture — **IDLE CUT-OFF (short final, landing assured)**
- Magnetos — **OFF**
- Fuel selector — **OFF**
- Master switch — **OFF**
- Door — **UNLATCH**
- Touchdown — **MINIMUM speed, nose slightly high, then full aft**
- Aircraft — **EVACUATE when stopped**

Prototype content. Verify against the operator's approved checklist and the POH/AFM before any use. AeroE Checklist version 1.7 (build 849ad9ab) · 2026-11-07